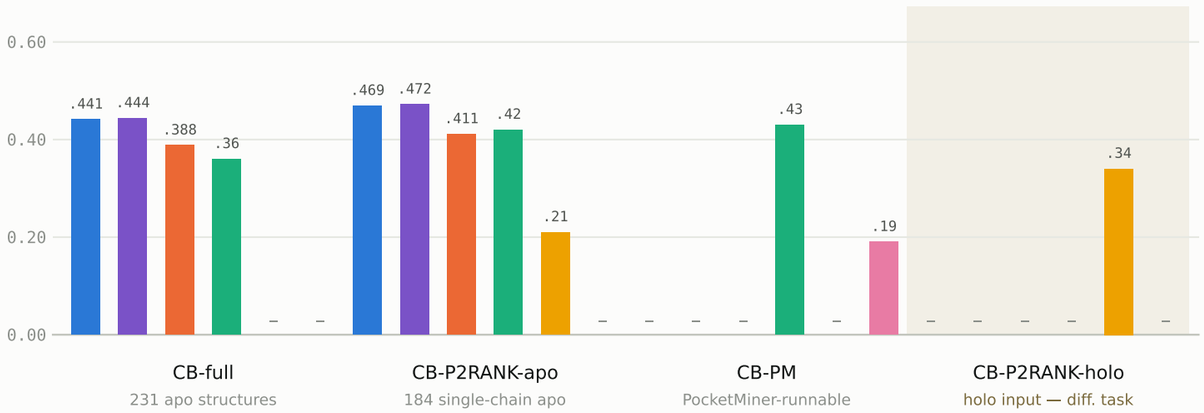
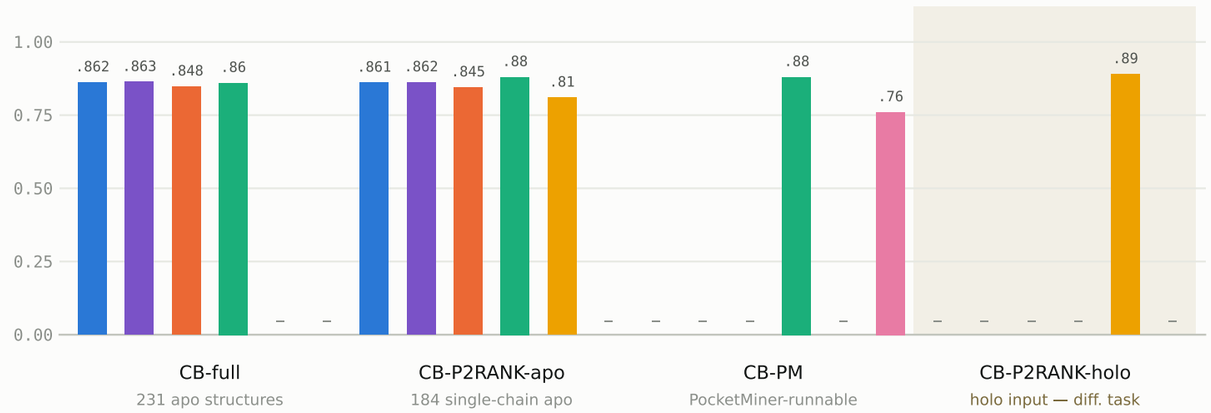
Per-residue cryptic site prediction on CryptoBench

All metrics per structure then averaged (CryptoBench Protein.py convention). Test split: 231 apo structures, 76,636 residues, 5.28 % positive.

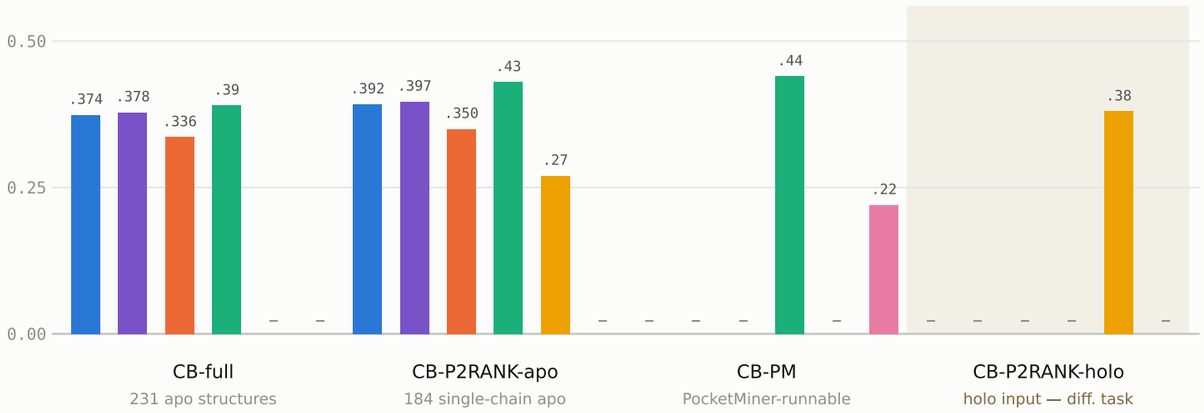
AUPRC



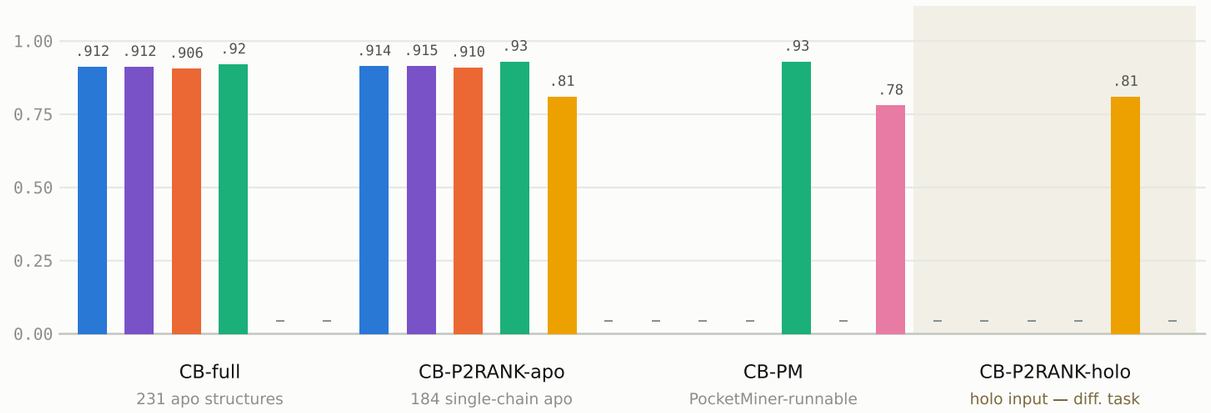
AUC



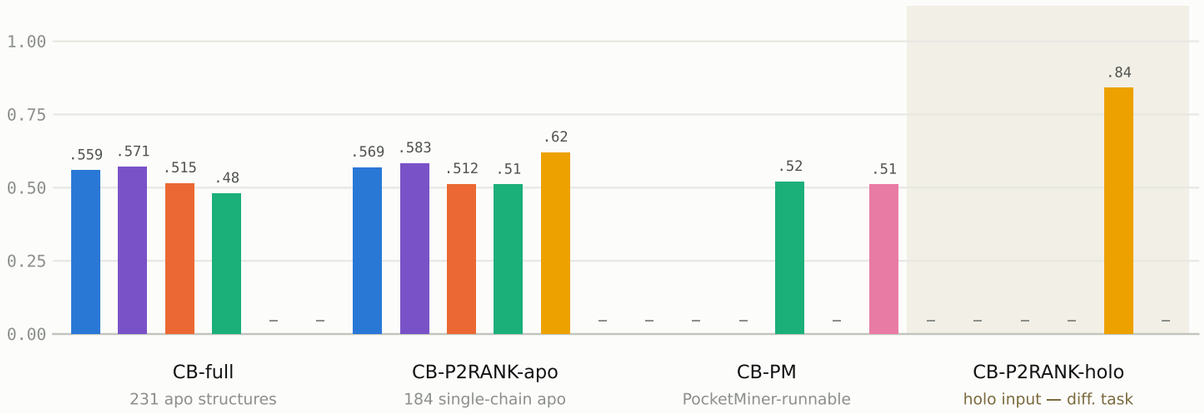
MCC



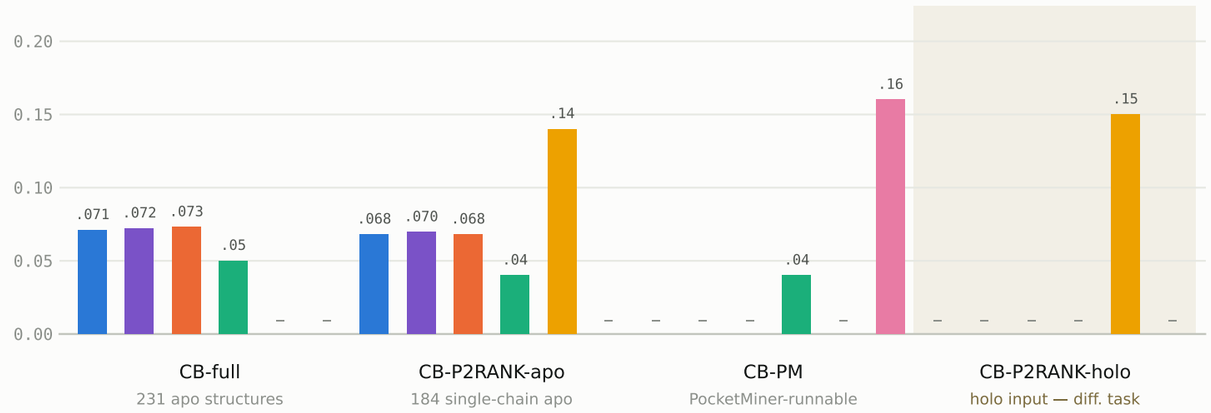
F1 (weighted)



TPR



FPR ↓ lower is better



- TabICLv2 + ESM, zero-shot (this work)
- TabICLv2 + ESM, fine-tuned (this work)
- TabICLv2, no pLM (this work)
- pLM-NN (benchmark)
- P2Rank (benchmark)
- PocketMiner (benchmark)

– method not evaluated on that subset.
CB-PM is defined by which structures PocketMiner failed on, so it cannot be reconstructed from the shipped files.

Shaded group: P2Rank on holo (ligand-bound) input — a different task, shown for context, not a fifth column of apo results.

TPR / FPR / MCC read each method at its own threshold (pLM-NN 0.95, PocketMiner 0.7, P2Rank binary, ours frozen on train folds at FPR 0.05). AUC and AUPRC are threshold-free.

On F1
The published F1 is a weighted average over BOTH classes, so at 5.28 % positives it is governed by the 95 % of residues that are trivially negative. Every pLM-based row lands at 0.92–0.93; the column separates the structure-only tools and nothing else.

Our positive-class F1 — the number that carries information — is 0.379 (117 feat) and 0.345 (53 feat) on CB-full. It is NOT comparable to the published column and is deliberately not plotted beside it.

Reproduce every value above with:
    python scripts/f1_by_subset.py
Fine-tuned bars: fine_tuned_tabicl/ (deploy/finetune_modal.py); its zero-shot control reproduces the blue bars to 3 dp.